

# Leave-One-Out Meta-Analysis

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## Introduction

Leave-one-out (LOO) meta-analysis systematically removes each study and re-computes the pooled estimate using the remaining  $k - 1$  studies. The resulting per-removal estimates show how much each study drives the overall conclusion. Studies whose removal substantially changes the pooled effect — magnitude, significance, or both — are influential and warrant careful interpretation. LOO is a routine sensitivity analysis in modern meta-analysis reporting.

## Prerequisites

A working understanding of pooled meta-analysis (fixed or random effects) and the concept of influence diagnostics from regression analysis.

## Theory

For each study  $i = 1, \dots, k$ , compute  $\hat{\theta}_{(-i)}$  by pooling the other  $k - 1$  studies under the same meta-analysis model. Plot  $\hat{\theta}_{(-i)}$  with its confidence interval across  $i$ ; large deviations from the full-model pooled estimate indicate influential studies.

LOO complements other influence measures: Cook's distance and DFBETAS quantify how much each study perturbs the overall fit, and Baujat plots show each study's contribution to heterogeneity ( $Q$ ) versus its contribution to the pooled estimate. Used together, these diagnostics provide a comprehensive picture of which studies the conclusion depends on.

## Assumptions

Same as the underlying random-effects meta-analysis; each leave-one-out fit uses the same model and method as the full pooling.

## R Implementation

```
library(metafor)

set.seed(2026)
k <- 10
yi <- c(0.35, 0.42, 0.25, 0.55, 0.28, 0.39, 0.95, 0.31, 0.44, 0.36)
vi <- c(0.02, 0.03, 0.015, 0.02, 0.025, 0.018, 0.01, 0.04, 0.02, 0.022)

res <- rma(yi, vi, method = "REML")
loo <- leavelout(res)

print(loo[, c("estimate", "ci.lb", "ci.ub", "I2")])

# Influence diagnostics: Cook's distance, DFBETAS
inf <- influence(res)
print(inf$inf)
```

## Output & Results

`leave1out()` returns a per-study table of the recomputed pooled estimate, confidence interval, and  $I^2$  when each study is omitted. `influence()` provides Cook’s distance, DFBETAS, hat values, and other diagnostics computed jointly.

## Interpretation

A reporting sentence: “Leave-one-out analysis revealed study 7 (SMD = 0.95, the largest single estimate) as highly influential; removing it reduced the pooled SMD from 0.41 to 0.35 and shrunk  $I^2$  from 38 % to 18 %. Conclusions are robust to removal of any other study; the sensitivity analysis without study 7 is reported as supplementary.” Always describe what an influential study removal does to both effect size and heterogeneity.

## Practical Tips

- Pair LOO with Cook’s distance and DFBETAS from `metafor::influence()`; the combination flags both effect-size influence and overall fit influence.
- If removing any single study crosses the significance line, the conclusion is fragile and should be reported as such.
- LOO is descriptive — it does not correct any bias. Use it to inform readers about robustness, not to “clean up” the meta-analysis post-hoc.
- Report LOO results only if at least one study is markedly influential; otherwise the table conveys no useful information beyond the headline pooled estimate.
- Combine with Baujat plots (`metafor::baujat()`) for a richer visualisation: studies in the upper-right corner contribute disproportionately to both heterogeneity and effect.
- For meta-analyses with fewer than 5 studies, LOO is unstable and largely uninformative; report it but interpret cautiously.

## R Packages Used

`metafor::leave1out()` for the canonical LOO output; `metafor::influence()` for Cook’s distance, DFBETAS, and hat values; `metafor::baujat()` for Baujat plots; `meta` for analogous sensitivity tools in an alternative interface; `dmetar::infA()` for combined influence and outlier diagnostics in an integrated workflow.

## For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

## See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis